IEEE VTC-Spring 2027: The 4rd Workshop on
Synergizing Digital Twins and Pervasive Intelligence for Next-Generation IoV
Call for Papers

As next-generation Internet of Vehicles (IoV) systems evolve toward increasingly autonomous, interconnected, and intelligence-driven operation, their communication, computing, and control infrastructures must support dense connectivity, high mobility, heterogeneous services, and stringent latency and reliability requirements. Digital twins (DTs) provide continuously updated virtual representations of vehicles, networks, and transportation environments, while pervasive artificial intelligence (AI) distributes learning, inference, prediction, and decision-making capabilities across vehicles, roadside units, edge servers, and cloud platforms. Their integration establishes a bidirectional physical–virtual intelligence loop in which DTs support data-driven modeling and optimization, whereas AI enhances state estimation, prediction, adaptation, and autonomous control. However, maintaining accurate physical–virtual correspondence remains challenging under rapid mobility, time-varying wireless channels, dynamic traffic conditions, and evolving network topologies, which can result in delayed updates, DT desynchronization, and physical–virtual domain gaps. Meanwhile, large-scale DT maintenance and pervasive AI execution introduce substantial communication and computation overhead, potentially causing congestion, processing delays, reduced DT freshness, and degraded decision quality. These challenges motivate scalable DT–AI architectures, adaptive synchronization, domain adaptation, edge–cloud collaboration, task partitioning, cooperative processing, computation offloading, distributed inference, and joint communication–computation resource orchestration..

This workshop invites high-quality original research contributions that explore theories, architectures, algorithms, and experimental systems for synergizing digital twins and pervasive intelligence in next-generation IoT/IoV. Both theoretical advances and practical insights from testbeds, prototypes, and field trials are encouraged.

Topics of Interest (including but not limited to)	Topics of Interest (including but not limited to)
• DT architectures and modeling techniques for IoV systems • AI-enhanced digital twins for prediction, optimization, and autonomous control • Physical-virtual synchronization, DT fidelity, and state reconstruction • Domain adaptation, continual learning, and cross-domain knowledge transfer • Pervasive intelligence in self-organizing and autonomous vehicular systems • Edge-cloud collaboration for DT-enabled intelligence • Distributed and cooperative learning for DT-AI integration • Resource management, routing, scheduling, and network slicing enabled by DT-AI	• Communication-computation-aware DT-AI co-design • Adaptive task partitioning and cooperative processing • Intelligent computation offloading and distributed inference • Big-data management and real-time analytics for large-scale DTs • DT-AI-enabled intelligent transportation and smart infrastructure • UAV/drone-assisted vehicular sensing and monitoring using DT and AI • Interoperability, evaluation, and standardization for DT-AI-enabled 6G IoV • Experimental platforms, prototypes, testbeds, and field trials

Paper submission: Papers must be formatted in the standard IEEE two-column format that is used by the VTC-Spring 2026 main conference, and must not exceed six pages in length (including references).

Workshop Organization Committee

General Co-chairs
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Important Dates:
Paper Submission Deadline: 20 Dec. 2026
Paper Acceptance Notification: 1 Feb. 2027
Final Paper Submission: TBD